
PERSONAL SUMMARY

2nd year Materials Engineering student and aspiring aerospace robotics engineer. Chassis & Communications Lead on McMaster's Mars Rover Team and undergraduate robotics researcher at McMaster RMAL. Skilled across CAD/FEA, ROS2 perception, and fabrication.

EDUCATION

McMaster University*Candidate for BEng in Materials Engineering*

Hamilton, ON

Sep 2025 – Apr 2029

University of Waterloo*Quantum School for Young Students, Institute of Quantum Computing*

Waterloo, ON

Jul 2025 – Aug 2025

EXPERIENCE

Chassis & Communication Systems Lead, Mechanical Subteam

August 2026 – Current

McMaster Mars Rover Team

Hamilton, ON

- Led entire design and deployment of the ground basestation, antenna, and pan-tilt camera assembly, taking the communications subsystem from concept to field operation.
- Produced engineering drawings and machined structural components, conducting FEA to validate mechanical integrity under rover operating loads.
- Promoted from sole contributor to sub-team lead after independently delivering the subsystem; now direct a team of 5, owning technical decisions, integration, mentorship, and sub-team budgets and timelines.

Undergraduate Robotics Researcher

Aug 2025 – Current

McMaster Robotics and Manufacturing Automation Laboratory

Hamilton, ON

- Developed a novel grasp-planning algorithm for agricultural robotics in MATLAB, using multi-height cross-sectional slicing and fruit-specific deformability modeling to generate grasps for 5 fruit geometries from 3D mesh input.
- Built an end-to-end validation preception pipeline in ROS 2 (Jazzy) and Gazebo Harmonic to test the algorithm on simulated depth-sensor data, integrating a 3-fingered depth-camera gripper for automated point-cloud capture, segmentation, and shape completion across dozens of programmatic viewpoints.
- Validated the perception-to-grasp system over 108 trials spanning 3 sensor-noise levels, achieving <0.5 mm diameter-estimation bias and <2.5 mm MAE at realistic noise, with an automated quality gate filtering ~30% of degenerate viewpoints.
- Bridged a CAD-mesh-based planning algorithm to realistic sensor input by developing a partial-view-to-mesh reconstruction pipeline (depth capture → segmentation → sphere fitting → symmetry completion → watertight meshing), enabling the planner to operate on noisy, occluded point clouds instead of idealized geometry.

Undergraduate Researcher

Aug 2025 – Current

McMaster Health Sciences Lupus Clinic and Aletheia

Hamilton, ON

- Founded a computational immunogenetics initiative combining gene expression analysis and ML to develop diagnostic biomarkers for autoimmune disease, in collaboration with doctors at McMaster, UofT, and McGill.
- Published bioinformatics research analyzing 54,675 gene probes across three immune cell types to identify a novel 5-gene expression signature for systemic lupus erythematosus, and diagnostic criteria for a potential single blood-test tool.

Team Member, Design & Build

Sep 2024 – Dec 2024

FIRST Robotics Team 1360

Ontario, ON

- Contributed to the mechanical design and fabrication of a competition robot using 3D printing, lathe, and machining.
- Prototyped and iterated components within a build subteam under competition deadlines.

Space Camp — Wings Graduate

Jul 2019 – Sep 2019

U.S. Space & Rocket Center

Huntsville, AL

- Completed astronaut-style training simulations and mission scenarios emphasizing teamwork and communication.
- Worked in teams on guided engineering and spaceflight challenges across science and technology modules.

Member, Design & Build — Project Lead

Sep 2018 – Apr 2019

FIRST Robotics Team 3436

Ontario, ON

- Led a design-and-build effort as project lead, coordinating fabrication tasks including 3D printing.
- Built and tested robot components for regional FIRST competition.

PUBLICATIONS

LeBlanc, M. "Analysis of CD16 +/- Monocytes and CD4 + T Cells to Identify Novel Gene Signatures and Develop a Diagnostic Tool for SLE." *International Journal of High School Research*, 2025

LeBlanc, M. "Carbon Capture Depletion Using Catalyzed Photosynthesis to Promote Crop Growth." *Canadian Science Fair Journal*, 2023.

AWARDS, FELLOWSHIPS, AND GRANTS

ISEF 4th Place Grand Award in Biomedical and Health Sciences (2024)

Junior Astronauts Upswing Academy — Canadian Space Agency & Government of Canada (2020)

Queen's Platinum Jubilee Medal & MP Recognition Certificate — MP Anita Anand (2022)

Schulich Leader Nominee — top STEM student in graduating class (2025)

Ingenious+ Youth Innovation Challenge Regional Award — Rideau Hall Foundation (2025, 2026)

Hamilton Chamber of Commerce Innovation Award — First Place (2023)

Sanofi Biogenius Canada Award (2024)

FIRST Lego League Champions Award — Peel Qualifying Tournament (2019)

PEO Bridge Building Competition — 3rd Place (Force), 5th Overall (2024)

John W. Howard Materials Research Award (2021)

Association for Iron and Steel Technology (AIST), Northern Chapter Award (2023)

ArcelorMittal Dofasco Environment Award (2023)

Canadian Nuclear Society Award — Intermediate, Golden Horseshoe Branch (2022)

CWSF 4 Awards (2 Bronze, 1 Merit, 1 Intact Climate Change Resilience Award) (2021–2024)

BASEF Best in Fair & Best Intermediate/Senior Project (2024)

BASEF 20 Awards (2019–2024)

Spirax-Sarco Scholarship for Future Innovators in Engineering — Girls SySTEM (2025)

Canadian Federation of University Women Award (2025)

ACSI Public Speaking Festival — 1st Place, “Girls in Robotics” (2019)

PROJECTS (SELECTED)

- Dual-GPS Antenna Mast — Mars Rover (URC 2026)**

Jan 2026 – Current

- Solo mechanical and materials design of a dual-GPS antenna mast: four antennas at a 1 m baseline on a hollow carbon-fibre mast with PLA brackets, validated with modal FEA.
- 2-DOF Pan-Tilt Camera System — Mars Rover (URC 2026)**

Jan 2026 – Current

- Solo design of a servo-actuated pan-tilt camera mount with a custom 1:2 spur gear train for 360° pan and a 608 bearing that resolved the prior mount’s gear failure; SolidWorks, PLA FDM.
- ROS2 Robot Simulation and Sensor Integration**

May 2026 – August 2026

- Built and simulated a differential-drive robot in ROS2; modeled geometry in URDF/Xacro, integrated LiDAR via TF transforms, and enabled keyboard teleoperation in Gazebo.
- EEG-Controlled Prosthetic Hand (Project AURA)**

May 2025 – Jun 2025

- Designed EEG-driven prosthetic with Arduino, Python signal processing, and 3D-printed ergonomic frame.
- Adaptive Robotic End Effector — Q-Arm Warehouse Automation**

Sep 2025 - Dec 2025

- Designed a sub-100 g 3D-printed gripper and modular Python control stack enabling a single-motor Q-arm to scan, pick, and pack objects of varying shape.

TECHNICAL SKILLS

Design/CAD: SolidWorks, FEA, PDM, AutoCAD, Inventor, Onshape, Revit, Autodesk 3ds Max

Robotics/Hardware: Raspberry Pi, ESP32, Arduino, 3D printing, machining

Programming/Tools: Python, Bash, MATLAB, ROS 2 (Jazzy), Gazebo Harmonic, RViz, URDF/Xacro, ros2_control, ros_gz.bridge, Open3D, NumPy/SciPy, Git, Linux (Ubuntu 24.04)

Data/Research: Excel, Minitab Statistical Software, NCBI GEO

3D Perception/Algorithms: Point cloud processing, RANSAC and geometric fitting, DBSCAN segmentation, Poisson surface reconstruction, depth-camera simulation, grasp planning

Other: LaTeX, Microsoft 365, Photoshop, Illustrator, Ansys Granta EduPack

PROFESSIONAL AFFILIATIONS

- Society for Science ISEF** — Alumnus
 - Girls SySTEM Mentorship Program** — Mentee
 - Sigma Xi, The Scientific Research Honor Society** — Member

May 2024 – Present

Sep 2022 – Sep 2025

Dec 2024 – Sep 2025

CERTIFICATIONS

Pilot Certificate — Basic Operations of RPAS (Remotely Piloted Aircraft System) — Transport Canada (2026)

Robotics Dev Fundamentals: Linux, C++, ROS2, micro-ROS — Robotisim (2025)

Python 3 for Robotics — The Construct Robotics Institute (2025)

Linux for Robotics — The Construct Robotics Institute (2025)

SOLIDWORKS Essentials Training — FEA Training Consulting Inc. (2025)

Networking Basics — Cisco Networking Academy (2025)

Digital4Sustainability Learning Path — United Nations System Staff College (2026)

Level 8 Piano, Practical & Theory, First Class Honors — The Royal Conservatory of Music (2022–2023)